

Time-Ordered Schur Cross-Gramians for Directional Hardy Fusion

Removing Radial Riesz Obliqueness

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July 2026

Abstract

RH-57 expressed directional Hardy energies through cross-Gramians built from invariant Riesz blocks. Its fixed radial projectors became severely oblique as the noise decreased. This paper replaces those projectors by a unitary ordered Schur basis and identifies exactly what that replacement gains and what it does not.

For each stable finite matrix $A = QTQ^*$, we prove two dual positive packet identities for every orthogonal coordinate partition of its triangular Schur form. The input identity decomposes the complete response into orthogonal initial Schur packets through the observability Gramian; the output identity decomposes the controllability Gramian into observed Schur state blocks. Both reconstruct the same Hardy energy and admit coherence-weighted square-sum bounds. We also prove a reverse-order cross-Stein recursion for the blocks of the controllability Gramian. A block-power formula bounds every diagonal Stein inverse without forming a Kronecker matrix.

The deterministic five-scale binary64 audit is sharply positive at the exact packet level. At the smallest stored scale, the left/right input-packet coherence uppers are 1.5527 and 1.9009, while the output-block uppers are 1.4682 and 1.7612. The corresponding RH-57 radial Riesz uppers were 81.68 and 672.41. Every diagonal Schur block has eight-step norm below 0.290, and the largest observed block-Stein gain is below 2.80.

There is also a new negative result. If every feed-forward Schur path is replaced by spectral/Frobenius norms and summed absolutely, the sufficient upper grows to 1922.4 and 380.4. Thus unitary Schur coordinates remove Riesz obliqueness, but a scalar absolute-path proof discards the packet orthogonality that made the decomposition useful. A 256-bit Arb calculation certifies the formulas on a two-scalar-block model only. No dyadically uniform physical packet theorem, Stage A1 closure, arithmetic trace formula, or Hilbert–Polya conclusion is claimed.

Keywords: Schur form; cross-Gramian; Stein equation; Hardy energy; Haar channel; nonnormal operator; small noise.

MSC 2020: 47A10; 47B65; 37D25; 37M25; 65F35.

1 Introduction

The intrinsic small-noise program has reduced its remaining analytic premise to two directional Hilbert–Schmidt Hardy energies. RH-48–RH-50 derived the Schur-complement and time-domain reductions; RH-51–RH-56 supplied finite Stein tails, residue and cutoff transfer, and a quantitative obstruction to a global strong-space argument [4–11, 13]. RH-57 then proved an exact Riesz cross-Gramian identity and tested fixed radial invariant blocks [12].

The RH-57 result exposed a precise mismatch. The exact all-column Hardy energies remained below 1.77 on the stored dense levels, but fixed radial Riesz projectors reached norms

above 2.5×10^3 . Their individual block responses became large and cancelled only after aggregation. Consequently, an absolute invariant-block budget was not a credible uniform proof route.

The Schur form is the natural next coordinate system [1, 2]. It replaces oblique spectral projections by a unitary basis while retaining a time ordering: outer spectral packets feed inner packets through the strictly upper block part of T . This paper answers two separate questions.

1. Does the unitary packet decomposition itself remain quantitatively mild on the growing finite family?
2. Can that mild decomposition be proved by taking absolute norms along every upper-triangular feed-forward path?

The audit answers yes to the first and no to the second. This distinction is the main route information supplied by RH-58.

Contributions and boundary

1. We prove dual input-packet and output-state Gram identities in an arbitrary unitary Schur partition. No diagonalizability is required.
2. We prove the reverse cross-Stein block recursion induced by upper triangularity.
3. We derive a computable block-power upper for each Stein inverse and a rigorous scalar absolute-path majorant.
4. We audit both exact packet Grams and the scalar path majorant on five all-column folded-Gaussian levels, with all primal, dual, Schur, and recursion residuals stored.

The finite-dimensional identities are exact. The ordered production Schur forms are binary64 and noise-dependent. Nothing below proves continuum regularity of their packet subspaces or a uniform small-noise Hardy budget.

2 Directional Hardy system

Both RH-50 directional energies can be written as

$$\mathcal{E}(r)^2 = \sum_{m \geq 0} r^{-2m} \|Y N^m X\|_{\mathfrak{S}_2}^2 = \sum_{m \geq 0} \|Y A^m X\|_{\mathfrak{S}_2}^2, \quad A = r^{-1} N, \text{quad spr}(A) < 1. \quad (1)$$

For the left triple,

$$X = Q_f U B / \|B\|_{\mathfrak{S}_2}, \quad Y = U^*;$$

after adjoining the right triple,

$$X = C^* / \|C\|_{\mathfrak{S}_2}, \quad Y = Q_c^*.$$

The controllability and observability Gramians are

$$G - A G A^* = X X^*, \quad G = \sum_{m \geq 0} A^m X X^* (A^*)^m, \quad (2)$$

$$O - A^* O A = Y^* Y, \quad O = \sum_{m \geq 0} (A^*)^m Y^* Y A^m. \quad (3)$$

They give the primal–dual trace identity

$$\mathcal{E}(r)^2 = \text{tr}(Y G Y^*) = \text{tr}(X^* O X). \quad (4)$$

Choose a complex Schur form ordered from the central disk toward the outer cloud,

$$A = QTQ^*, \quad Q^*Q = I, \quad T = \begin{pmatrix} D_1 & T_{12} & \cdots & T_{1J} \\ 0 & D_2 & \cdots & T_{2J} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & D_J \end{pmatrix}. \quad (5)$$

Let E_j be the orthogonal coordinate projections onto the Schur blocks and put

$$\hat{X} = Q^*X, \quad \hat{Y} = YQ, \quad X_j = E_j\hat{X}, \quad Y_j = \hat{Y}E_j. \quad (6)$$

Unlike the Riesz projections of RH-57, every E_j has norm one. It is not invariant under T ; its later motion is recorded by the upper-triangular couplings in (5).

3 Dual Schur packet Gramians

Write $\hat{G} = Q^*GQ$ and $\hat{O} = Q^*OQ$. Their block entries are $G_{ij} = E_i\hat{G}E_j$ and $O_{ij} = E_i\hat{O}E_j$.

Theorem 3.1 (Dual unitary Schur packet identities). *Let A be a finite matrix with $\text{spr}(A) < 1$, and let (5) be any unitary Schur partition. Define the input-packet matrix*

$$K_{ij}^{\text{in}} := \text{tr}(X_j^*\hat{O}X_i) = \sum_{m \geq 0} \text{tr}(\hat{Y}T^m X_i X_j^* (T^*)^m \hat{Y}^*), \quad (7)$$

and the output-state matrix

$$K_{ij}^{\text{out}} := \text{tr}(Y_i G_{ij} Y_j^*). \quad (8)$$

Then both K^{in} and K^{out} are positive semidefinite and

$$\boxed{\mathcal{E}(r)^2 = \mathbf{1}^* K^{\text{in}} \mathbf{1} = \mathbf{1}^* K^{\text{out}} \mathbf{1}.} \quad (9)$$

For either Gram matrix K , let $e_j = K_{jj}^{1/2}$ and $C_{ij} = K_{ij}/(e_i e_j)$ on the nonzero diagonal indices. Then

$$\mathcal{E}(r)^2 \leq \lambda_{\max}(C) \sum_j e_j^2. \quad (10)$$

No eigenvector condition number or oblique spectral projection appears.

Proof. For the input identity, define the complete packet response $F_i = (\hat{Y}T^m X_i)_{m \geq 0}$. Equation (3) gives $K_{ij}^{\text{in}} = \langle F_i, F_j \rangle_{\ell^2(\mathfrak{S}_2)}$, with the trace inner product taken linear in the first argument. Hence K^{in} is a Gram matrix. Since $\sum_i X_i = \hat{X}$, summing its entries gives (4).

For the output identity, choose L with $\hat{G} = LL^*$. The vectors $Y_i E_i L$ have Gram matrix (8). Summing them gives $\hat{Y}L$, so their total quadratic form is $\text{tr}(\hat{Y}\hat{G}\hat{Y}^*) = \mathcal{E}(r)^2$. Finally, writing $K = DCD$ with $D = \text{diag}(e_j)$ gives $\mathbf{1}^* K \mathbf{1} = e^* C e \leq \lambda_{\max}(C) e^* e$. $\square$

Remark 3.2 (Two different packet views). K^{in} groups the initial source by orthogonal Schur coordinates and then propagates each packet for all time. K^{out} groups the already accumulated controllability Gramian by the state coordinate observed at the end. Their entries differ, but their sums are the same Hardy energy. This dual agreement is a useful numerical consistency check and a choice of two possible analytic interfaces.

4 Time-ordered cross-Stein recursion

In Schur coordinates, (2) becomes

$$\widehat{G} - T\widehat{G}T^* = \widehat{X}\widehat{X}^*. \quad (11)$$

Upper triangularity orders its block equations from the outer cloud back to the central disk.

Theorem 4.1 (Reverse cross-Stein recursion). *For $1 \leq i, j \leq J$, define*

$$\mathcal{L}_{ij}(Z) = Z - D_i Z D_j^*. \quad (12)$$

Then $\mathcal{L}_{ij}$ is invertible and

$$G_{ij} = \mathcal{L}_{ij}^{-1} \left(X_i X_j^* + \sum_{\substack{a \geq i, b \geq j \\ (a,b) \neq (i,j)}} T_{ia} G_{ab} T_{jb}^* \right). \quad (13)$$

The blocks are therefore determined exactly by decreasing i and decreasing j .

Proof. The (i, j) block of $T\widehat{G}T^*$ is

$$\sum_{a \geq i} \sum_{b \geq j} T_{ia} G_{ab} T_{jb}^*.$$

Move the $(a, b) = (i, j)$ term to the left to obtain (13). Since the spectra of every D_i lie in the unit disk, the eigenvalues of $Z \mapsto D_i Z D_j^*$ have modulus below one. Thus $\mathcal{L}_{ij}$ is invertible. Every remaining pair (a, b) is later than (i, j) in the stated reverse order. $\square$

5 Block-power gains and an absolute-path majorant

The recursion is exact but still matrix-valued. A natural sufficient proof replaces each term by a Schatten norm. The first ingredient is a finite block-power bound for $\mathcal{L}_{ij}^{-1}$.

Proposition 5.1 (Block-power Stein gain). *Fix $M \geq 1$ and assume*

$$q_i := \|D_i^M\|_2 < 1 \quad (1 \leq i \leq J). \quad (14)$$

Then, on the Hilbert–Schmidt class,

$$\left\| \mathcal{L}_{ij}^{-1} \right\|_{\mathfrak{S}_2 \rightarrow \mathfrak{S}_2} \leq \left[\beta_{ij}^{(M)} := \frac{\sum_{s=0}^{M-1} \|D_i^s\|_2 \|D_j^s\|_2}{1 - q_i q_j} \right]. \quad (15)$$

Proof. The inverse is the convergent series

$$\mathcal{L}_{ij}^{-1}(R) = \sum_{m \geq 0} D_i^m R (D_j^*)^m.$$

Write $m = \ell M + s$, $0 \leq s < M$. Submultiplicativity gives

$$\|D_i^m R (D_j^*)^m\|_{\mathfrak{S}_2} \leq \|D_i^s\|_2 \|D_j^s\|_2 (q_i q_j)^\ell \|R\|_{\mathfrak{S}_2}.$$

Sum first over ℓ and then over s . $\square$

Put

$$\tau_{ia} = \|T_{ia}\|_2, \quad (16)$$

$$\xi_{ij} = \|X_i X_j^*\|_{\mathfrak{S}_2}, \quad (17)$$

$$\omega_{ij} = \|Y_j^* Y_i\|_{\mathfrak{S}_2}. \quad (18)$$

Starting from (J, J) and proceeding in reverse order, define

$$\Gamma_{ij} = \beta_{ij}^{(M)} \left(\xi_{ij} + \sum_{\substack{a \geq i, b \geq j \\ (a,b) \neq (i,j)}} \tau_{ia} \Gamma_{ab} \tau_{jb} \right). \quad (19)$$

Theorem 5.2 (Scalar absolute Schur-path upper). *Under (14),*

$$\|G_{ij}\|_{\mathfrak{S}_2} \leq \Gamma_{ij} \quad (20)$$

for every pair, and

$$\boxed{\mathcal{E}(r)^2 \leq \sum_{i,j=1}^J \omega_{ij} \Gamma_{ij}.} \quad (21)$$

Proof. Reverse induction in (13), the ideal inequality, and Proposition 5.1 give (20). Finally,

$$|\mathrm{tr}(Y_i G_{ij} Y_j^*)| = |\mathrm{tr}(Y_j^* Y_i G_{ij})| \leq \omega_{ij} \|G_{ij}\|_{\mathfrak{S}_2}.$$

Sum all block contributions absolutely. □

Remark 5.3 (Where cancellation is lost). The theorem is a valid sufficient upper, but it independently maximizes every Schur coupling and every cross-Gramian block. It preserves neither packet orthogonality nor the phase relations in K^{in} and K^{out} . The numerical audit below shows that this loss is decisive.

6 Five-scale deterministic audit

6.1 Construction

We use the RH-51 dense folded-Gaussian family with $N\sigma = 5.12$:

$$(\sigma, N) = (0.16, 32), (0.08, 64), (0.04, 128), (0.02, 256), (0.01, 512).$$

The intrinsic Perron and parity branches are removed exactly as in RH-50– RH-57. Both controllability and observability equations are solved densely, so every source and observation column is included.

The complex Schur form is ordered recursively by the same physical radial cuts used in RH-57,

$$0.15, \quad 0.35, \quad 0.55. \quad (22)$$

The central disk is selected first; each later sort acts only on the trailing Schur block. This preserves upper triangularity and keeps every previously selected packet fixed. Empty bands are omitted. The scalar majorant uses $M = 8$ in (15).

For each side and scale the archive stores the unitary and reconstruction defects, the primal–dual energy defect, both packet Grams, all recursion residuals, each diagonal block power norm, the observed and analytic Stein gains, and the RH-57 radial Riesz comparison.

6.2 Dual packet results

σ	exact $\mathcal{E}$		input-packet upper		output-block upper	
	left	right	left	right	left	right
0.16	0.9040	1.0026	0.9117	1.0068	0.9048	1.0061
0.08	1.1626	1.2653	1.1857	1.2905	1.1627	1.2656
0.04	1.3338	1.4845	1.3481	1.5143	1.3339	1.4853
0.02	1.4096	1.6340	1.4188	1.6992	1.4097	1.6352
0.01	1.4681	1.7603	1.5527	1.9009	1.4682	1.7612

Table 1: Exact all-column Hardy energies and the two unitary Schur coherence uppers. All quantities are binary64 diagnostics.

The largest input-packet upper is 1.90086 and the largest output-block upper is 1.76120. At $\sigma = 0.01$, RH-57's fixed radial Riesz coherence uppers were 81.68 and 672.41. Thus the large RH-57 values were not an intrinsic cost of separating radial spectral regions: they were a cost of enforcing an oblique invariant decomposition.

The observed finite-range power fits reinforce this distinction without proving an asymptotic law. The left/right exact energies have fitted powers 0.168 and 0.199; the input-packet uppers have powers 0.180 and 0.223; the output-block uppers have powers 0.167 and 0.199. None of these five-point fits establishes the RH-54 quarter-power budget.

6.3 Time ordering and the scalar-path wall

Every diagonal Schur block contracts over eight steps. Across all ten direction/scale cases,

$$\max_j \|D_j^8\|_2 < 0.290. \quad (23)$$

The largest empirical ratio $\|G_{ij}\|_{\mathfrak{S}_2} / \|\mathcal{L}_{ij}(G_{ij})\|_{\mathfrak{S}_2}$ is 2.793. The exact block recursion therefore remains modest in the stored range.

The scalar majorant behaves very differently.

σ	scalar path upper		RH-57 radial Riesz upper	
	left	right	left	right
0.16	1.59	1.78	1.92	2.40
0.08	4.41	4.15	2.99	5.90
0.04	22.39	14.26	6.11	12.73
0.02	137.26	71.18	40.94	180.24
0.01	1922.40	380.40	81.68	672.41

Table 2: Two sufficient bounds that take absolute norms too early. Neither table column is a lower bound on the true Hardy energy.

The fitted scalar-path powers are 2.54 and 1.96, far larger than the exact finite-range powers. At the smallest level the left path upper exceeds the exact energy by a factor about 1309. This is a no-go for the scalar majorant (19), not for Schur packets or for Stage A1 itself.

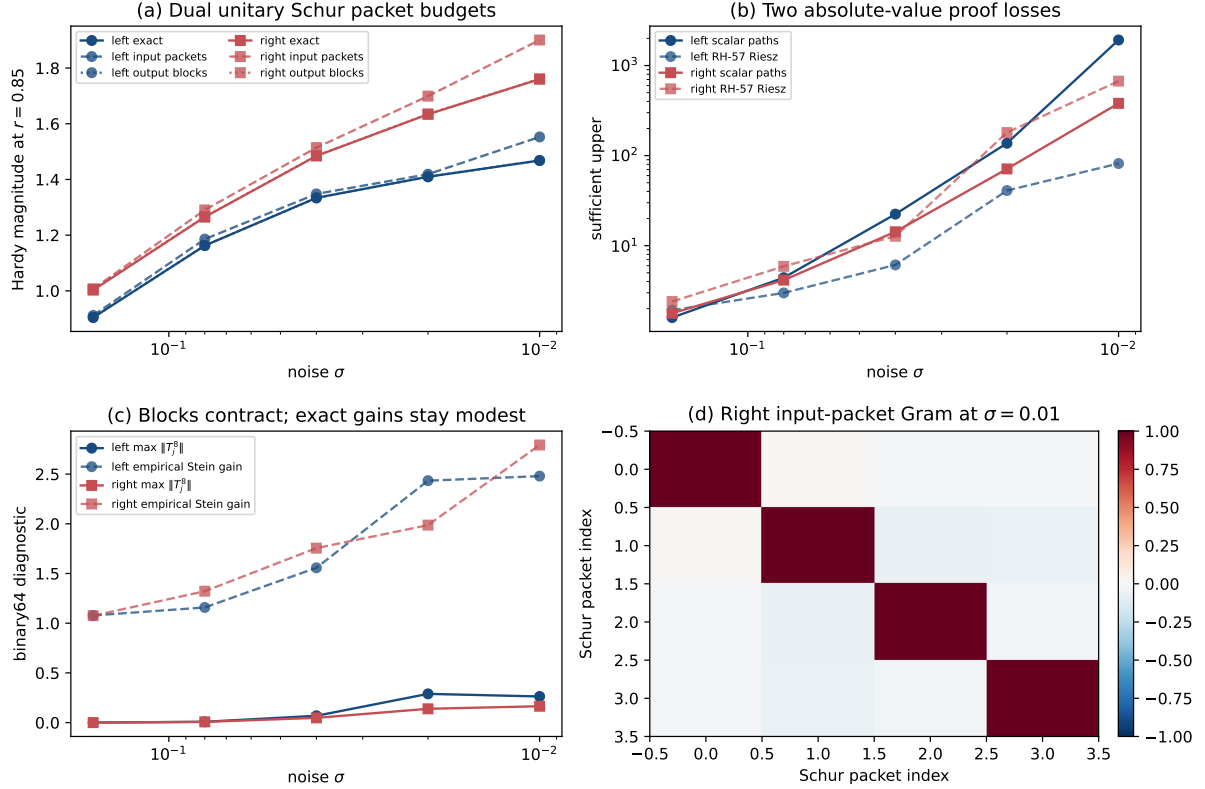


Figure 1: Unitary Schur audit. (a) Both exact packet budgets remain close to the all-column Hardy energies. (b) The RH-57 oblique Riesz route and the new scalar absolute-path route lose different cancellations. (c) All diagonal blocks contract over eight steps while exact recursion gains stay modest. (d) The right input-packet Gram at the smallest stored scale.

The maximum Schur reconstruction defect is 7.1×10^{-15} , the maximum cross-Stein recursion residual is 4.7×10^{-15} , and the maximum primal–dual trace discrepancy is 3.1×10^{-15} . These are binary64 consistency checks, not interval enclosures.

6.4 Outward-rounded model audit

A 256-bit Arb calculation [3] evaluates a real two-scalar-block upper-triangular model. It encloses the controllability and observability recursions, verifies that all six Stein residual balls contain zero, checks the primal–dual and dual packet identities, and certifies the $M = 2$ scalar path upper. No production folded-Gaussian Schur form is interval validated.

7 Program consequence

Fixed radial Riesz blocks

Rejected as a uniform absolute-overlap route by RH-57 obliqueness.

Unitary Schur packet coordinates

Quantitatively viable on the stored all-column levels. They remove the projector wall and give dual positive Gram interfaces.

Scalar absolute Schur paths

Rejected by Table 2. Block contraction alone does not prevent a combinatorial norm ledger from spending several powers of σ .

Remaining analytic target

Control the input packet Gram or output state-block Gram while preserving Hilbert-space square sums, phases, or an anisotropic block metric. Do not replace every feed-forward path by an independent worst-case norm.

Stage A1

Open. The present finite-dimensional theorem supplies a better coordinate system and a sharper target, not a uniform physical-family bound.

Stage A4

Still conditional on Stage A1; the RH-52–RH-55 interfaces remain intact.

The next gate is now narrower than after RH-57. One may seek a block observability inequality directly for K^{in} , or construct a positive block metric that sums Schur paths in square rather than absolutely. The stored data indicate that such a mechanism exists numerically, but they do not identify a continuum-stable Schur packet basis.

8 No arithmetic or Hilbert–Polya conclusion

Nothing here constructs a self-adjoint operator, a $T \log T$ counting law, a von Mangoldt or prime-power trace formula, or a zeta-zero identity. No Riemann-hypothesis conclusion is drawn. The independent twin-prime branch is not used.

9 Reproducibility

The archive contains the Schur algebra, tests, five-scale all-column results, the Arb model certificate, figures, dependency hashes, and publication artifacts. Principal commands are

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/run_schur_fusion_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/run_arb_schur_audit.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The dual packet identities, reverse recursion, block-power gain, and scalar path upper are analytic finite-dimensional statements. Production Schur forms and all fitted laws are binary64 diagnostics. Stage A1, unconditional intrinsic identification, and every arithmetic or Hilbert–Polya conclusion remain outside the claims.

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